European Network of
Transmission System Operators
For Electricity

DB Migration Guide

Energy Communication Platform
Version 4.16.0

Publication Date: 06.05.2025

Energy Communication Platform, version 4.16.0

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2. Executive Summary

This guide describes the process of data migration from an embedded database Apache Derby to an external database of ECP4 Endpoint.

This document is primarily targeted at Administrators.

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3. Revision History

Version	Date	Author	Description
4.6.0	22.11.2019	Anna Langrová	New template migration
4.6.1	18.03.2020	Anna Langrová	Version update
4.7.0	26.06.2020	Lukáš Voříšek	Version update
4.7.1	31.07.2020	Anna Langrová	Version update
4.7.2	16.11.2020	Anna Langrová	Version update
4.8.0	14.05.2021	Lukáš Voříšek	Version update
4.8.1	09.07.2021	Lukáš Voříšek	Version update
4.8.2	03.12.2021	Daniel Janota	Formatting
4.9.0	26.05.2022	Jiří Kakrda	Version update
4.10.0	07.11.2022	Daniel Janota, Jiří Kakrda	Hibernate, Oracle 12c and PostgreSQL deprecation
4.11.0	16.05.2023	Jiří Kakrda	Update of supported DBs
4.12.0	27.10.2023	Jiří Kakrda	Version update
4.14.0	10.06.2024	Jiří Kakrda	Artemis
4.15.0	21.10.2024	Jiří Kakrda	Version update
4.16.0	06.05.2025	Jiří Kakrda	SQL Server 2022, MySQL 8.4, PostgreSQL 16

4. ECP Endpoint database migration

This chapter describes the export of an Apache Derby database using the DBeaver tool. DBeaver is an open-source tool which is available for download at https://dbeaver.io/download/. The data import was tested for the MySQL database, using the MySQL Workbench tool (https://www.mysql.com/products/workbench/). However, the import procedure for other databases will be the same or very similar.

Supported external databases:

- MySQL 8.0
- MySQL 8.4
- Oracle 19c
- MSSQL Database Server 2019
- MSSQL Database Server 2022
- PostgreSQL Database server 13
- PostgreSQL Database server 16

4.1. Export data

Please note: ECP messages are not migrated to the external database. Only certificates and persistent application configuration are migrated. Please make sure that all messages are processed before the database migration. Unprocessed messages will be lost after migration. If the incoming traffic is stopped before migration, delete all message paths from the ECP Endpoint configuration. If the outgoing traffic is stopped, please stop all Business Applications connected to the ECP Endpoint (OPDM Client, EDX Toolbox...).

Follow the steps below in order to export all the required data from the embedded Derby database.

4.1.1. Connect to Embedded Database

1. Stop the ECP Endpoint
2. Copy the database from the ECP server to the location where the DBeaver client is installed. The Apache Derby database is stored in the folder db in the path specified in ECP configuration file ***ecp.properties*** – *dataDirectory* value.

   Default location of the embedded database:

   > Linux: /var/lib/ecp-endpoint/db

> Windows: <ecp-home>/db

3. Start DBeaver client, open ECP database

> Open the DBeaver application, go to the top menu, select File > New > Database connection > Expand the DBeaver folder > Database Connection > Expand the Derby folder > Embedded > Click on the “Browse” button and navigate to the “db” directory, fill in the credentials ecp/ecp-password > Next > Finish

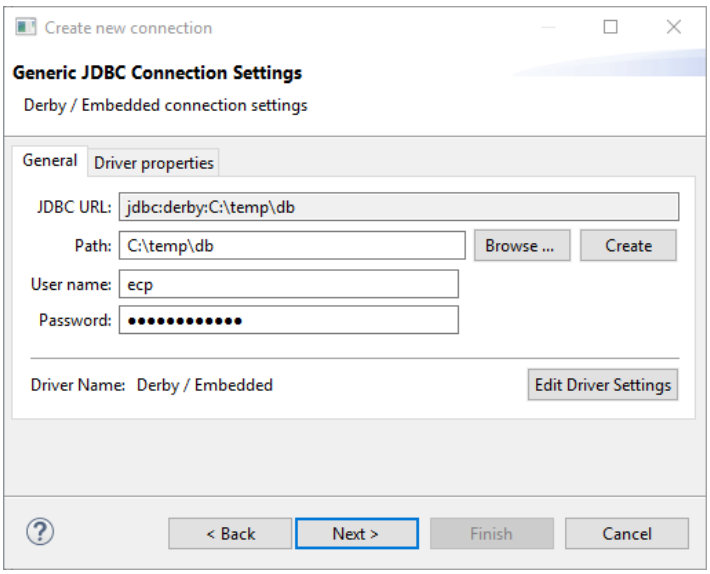

Figure 1: Create new connection

4. In the left menu, go to General > Connections > locate your ECP database (the default name is “Derby - db”) > ECP > Tables

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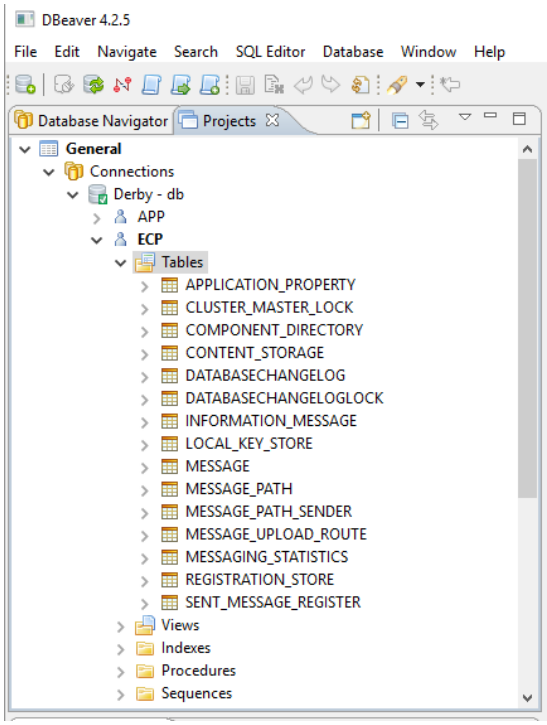

Figure 2: DBeaver

4.1.2. Export the Content of ECP Tables

Export of data from the tables below is required:

- APPLICATION_PROPERTY
- LOCAL_KEY_STORE
- MESSAGE_PATH
- MESSAGE_PATH_SENDER

1. Right click on the table name, click “Export Table Data”

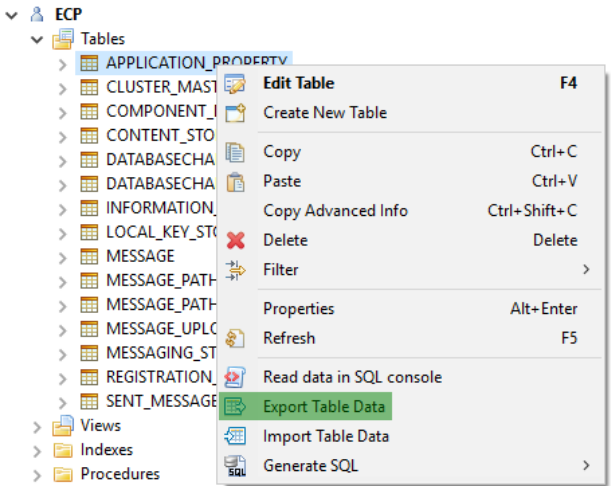

Figure 3: Export table data

2. Select “Export to SQL INSERT statements” -> Click Next > Next

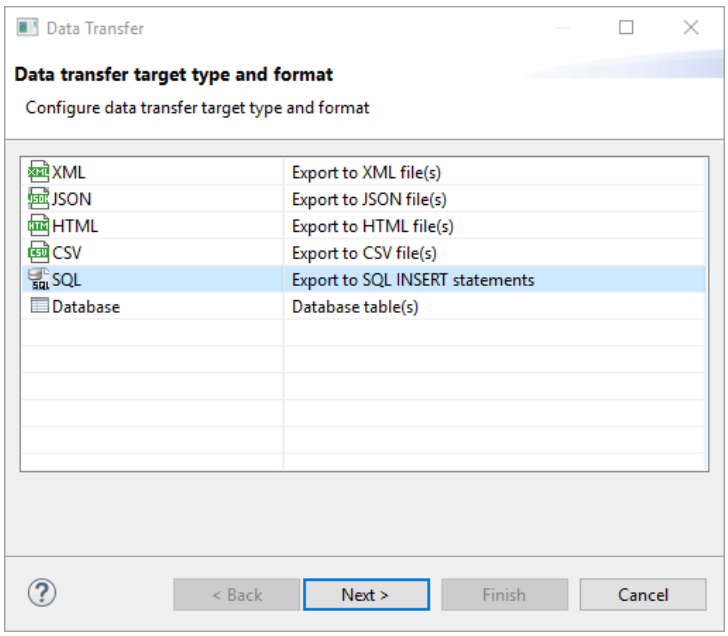

Figure 4: Data transfer - SQL

3. From the drop-down list Binaries, chose the “Inline” value and Encoding “Hex”, click Next

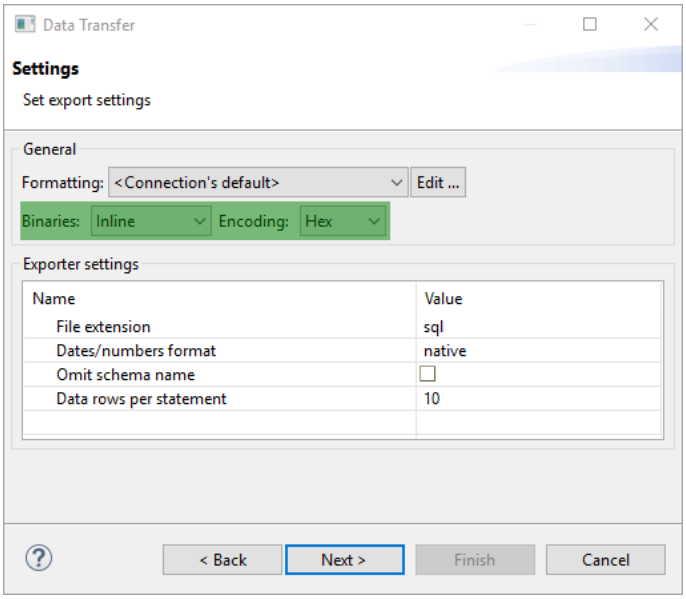

Figure 5: Data transfer - Settings

4. Fill in the Output folder for the SQL script. Make sure that UTF- 8 encoding is selected > Next > Finish.

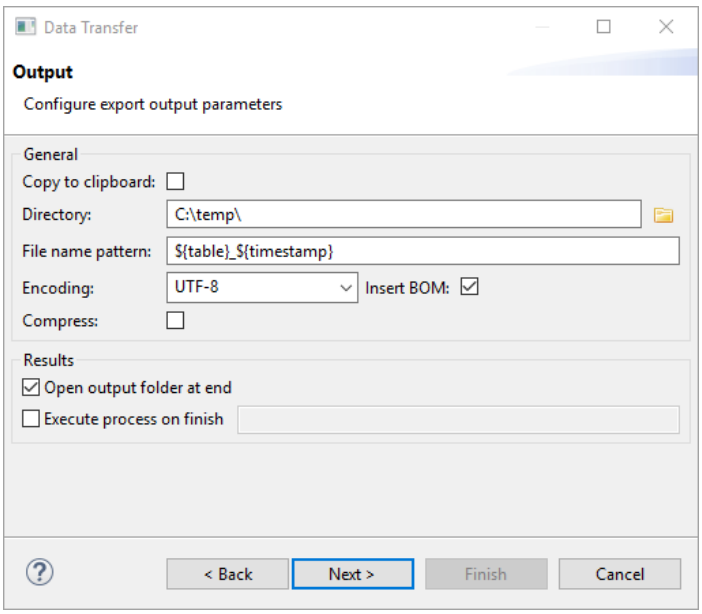

Figure 6: Data transfer - Output

5. Repeat the steps for all the requested tables:
   > APPLICATION_PROPERTY, LOCAL_KEY_STORE, MESSAGE_PATH, MESSAGE_PATH_SENDER.

4.2. Import data

4.2.1. Create Database Schema in the External Database

Before importing data, it is necessary to create an application schema and a user. Please, follow the *ECP Installation Guide*, chapter "*External Databases*".

Database tables will be created automatically after the DB connection string is changed to the new database in the ECP configuration file – as described in the following chapter.

4.2.2. Configure the ECP Application, Create Database Objects

The ECP Endpoint application needs to be configured for High Availability mode. Please, follow the *ECP Installation Guide*, chapter "*Parameters required for HA installation*". All parameters need to be configured.

1. Stop ECP Endpoint
2. Configure ECP Endpoint for HA mode (external database)
   - If your ECP Endpoint is installed on Windows, make sure that the character encoding for JDBC connection is set to UTF-8.
     - Example: jdbc:mysql://localhost:3306/ecp?autoReconnect=true?characterEncoding=UTF-8
3. Start ECP Endpoint
4. Check that ECP tables were created in your schemas. If the ECP tables were not created, check the application logs. See the pictures below – these tables should be created for the ECP Endpoint.

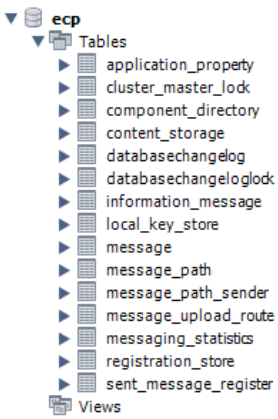

Figure 7: ECP - Tables

4.2.3. Import Data

For data import, use the SQL scripts generated in step "*Export content of ECP tables*".

1. Stop ECP Endpoint
2. Open your favourite database client
3. Connect to the ECP database
4. Open the SQL import scripts, change ECP schema name in the SQL scripts if needed.
5. Execute All SQL scripts. No error message should be returned.
6. Start ECP Endpoint
7. Verify ECP Endpoint functionality
   - Open the ECP Endpoint application > Go to the Settings tab > Scroll down to the Messaging Connectivity section, click the "Check Connectivity" button > Select your Endpoint code from the dropdown menu, fill in "test" as the Message type, click "Check" button. The status "CONNECTED" must be returned – as shown on the picture below.

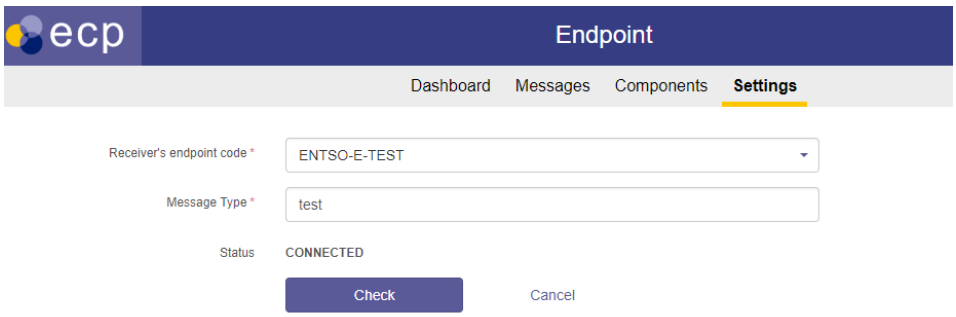

Figure 8: ECP - GUI

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